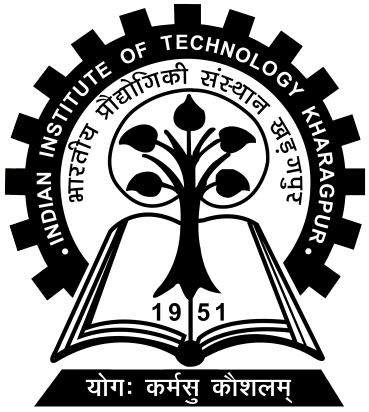
EXPERIMENT NO. 01—PN SEQUENCES



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TITLE: Design of Pseudo-Noise (PN) sequences using Linear Feedback Shift Register (LFSR) and applications of PN sequences.

PROBLEM STATEMENT: Design a 5-bit pseudo noise (PN) sequence generator using a linear feedback shift register (LFSR). In particular,

1. Demonstrate a simple construction of a PN sequence.
2. Now robustify the design by avoiding the 'lockout' condition (where all register states are set to '0').
3. Find the correlation function of the generated PN sequence by taking suitable shifted versions of the sequence.
4. Using simple arithmetic (modulo-2) operations, construct, generate and demonstrate a Gold code sequence.

KEY COMPONENTS: IC7486(XOR gate), IC7495(4-bit shift register), IC74164 (8-bit shift register), digital oscilloscope, function generator, breadboard, connecting wires, power supply.

BRIEF THEORY & BACKGROUND:
The objective of the experiment is to generate a pseudo-noise (PN) sequence and GOLD sequence using linear feedback shift registers.

A Linear Feedback Shift Register is a shift register where the input bit is done by some operation of the states (in our case XORing the previous states). This is mainly used to generate pseudo-noise sequence and Gold sequence [1].

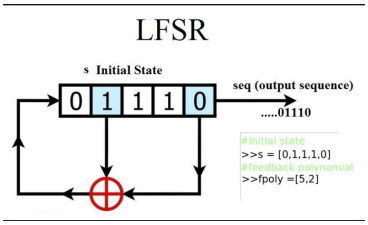


Figure 1: LFSR Diagram [2]

Pseudo Noise Sequences (PN Sequences) are sequences of 0 and 1 that look random at first glance, but after closely studyng its properties, they do not look random at all. In fact they can be generated using LFSr, by XORing the outputs (as per the primitive polynomial sequence) and getting the outputs [3].

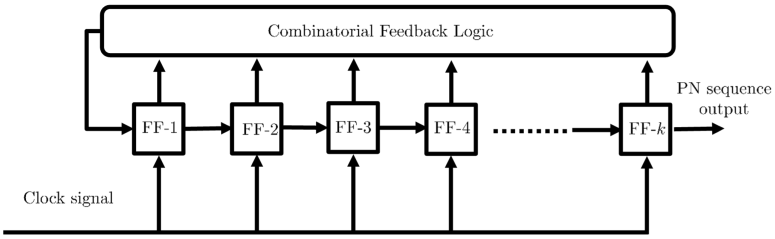


Figure 2: Block diagram of a typical PN sequence generator using an LFSR

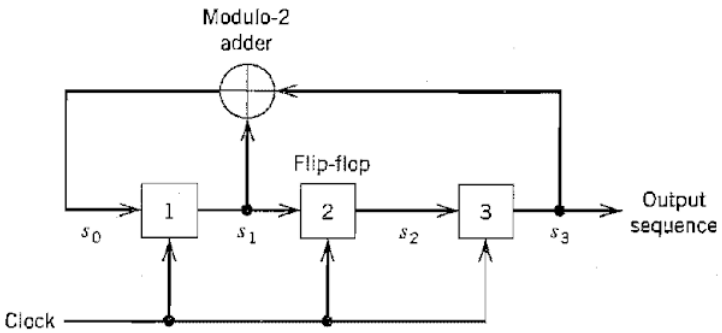


Figure 3: Block diagram of a typical PN sequence generator which has 3 memory elements [4]

Gold sequences are special binary sequences that are generated by XORing two PN sequences. To generate a Gold sequence, two same-length PN sequences have been chosen with identical periods and a common generating clock. Once the two PN sequences are generated, the final Gold sequence is generated by XORing the PN sequences component-wise [5].

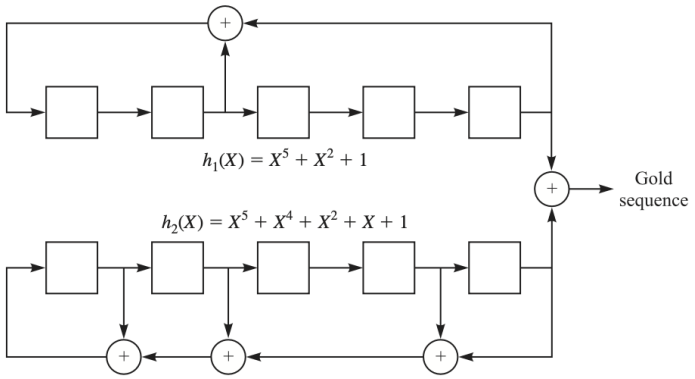


Figure 4: Block diagram of a Gold sequence generator using two PN Sequence generators and using XOR operation [6]

Properties of PN sequences:
The properties which make pseudorandom signal truly appears to be random are listed below:

1. **Balance Property:** In each period of a PN Sequence, the number of 1s is always one more than the number of 0s [7].

2. **Run Property:** A run in a PN sequence is a string of continuous values. Among the runs of 1s and 0s in each period of a maximum length sequence, one half the runs of each kind are of length one-fourth are of length two, one eighth are of length three, and so on as long as these fractions represent meaningful numbers of runs [7].

3. **Correlation Property:** Correlation is a measure of similarity between two sequences. When the two sequences compared are different it is the 'cross correlation' and when they are the same it is the 'autocorrelation' [7].

Applications of PN sequence in communication and signal processing systems:
PN sequences are widely used in generating test, measurement and calibration signals, training signals in communication systems, to scramble and descramble data in communication systems, and in uplink propagation (from mobile to basestation) so that basestation is able to distinguish users [8].

OUTCOMES:
Hardware:
The following is the plot of the PN Sequence generated in the lab. The sequence is as follows :

1100 1111 1000 1101 1101 0100 0010 010

Now there are 16 ones and 15 zeros, which matches the balance property.
There are total of 16 runs. Length 1 run = 8, length 2 run = 4, length 3 run = 2, length 4 run = 1 and length 5 run = 1. This aligns with the Run property.
To check the autocorrelation property, the sequence was plotted using autocorrelation function in MATLAB. Following is the plot:

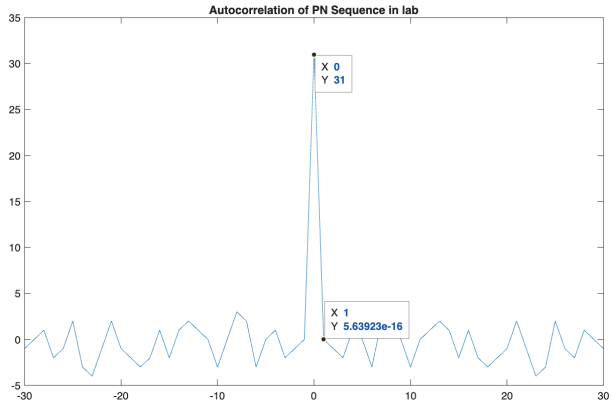


Figure 5: Autocorrelation plot of the PN Sequence. Note the drop around 0 which is almost 1/31 of the peak magnitude of autocorrelation plot.

The following is the PN Sequence generated in lab. Yellow plot shows the clock pulse and blue plot shows the PN Sequence plot.

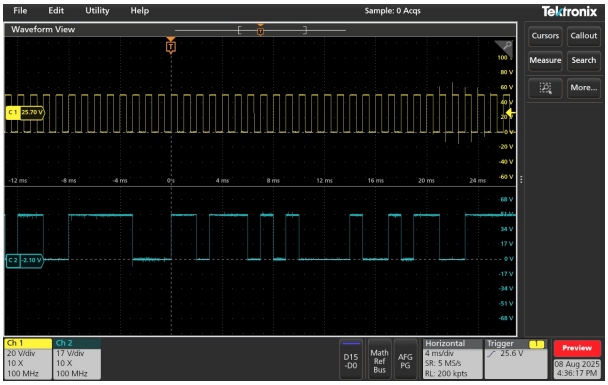


Figure 6: PN Sequence

The following is the Gold sequence plot done in Lab. Yellow plot shows the clock pulse and blue plot shows the Gold Sequence plot.N = 127 bit sequence was done in the lab, however not all bits are represented in the plot below.



Figure 7: Gold Sequence

Software:

1. Export the DSO data, i.e., time series in .csv format and plot it in MATLAB.
   The following figure plots the PN sequence, smoothed curve, clock sequence and the autocorrelation function using the data in the .csv file.

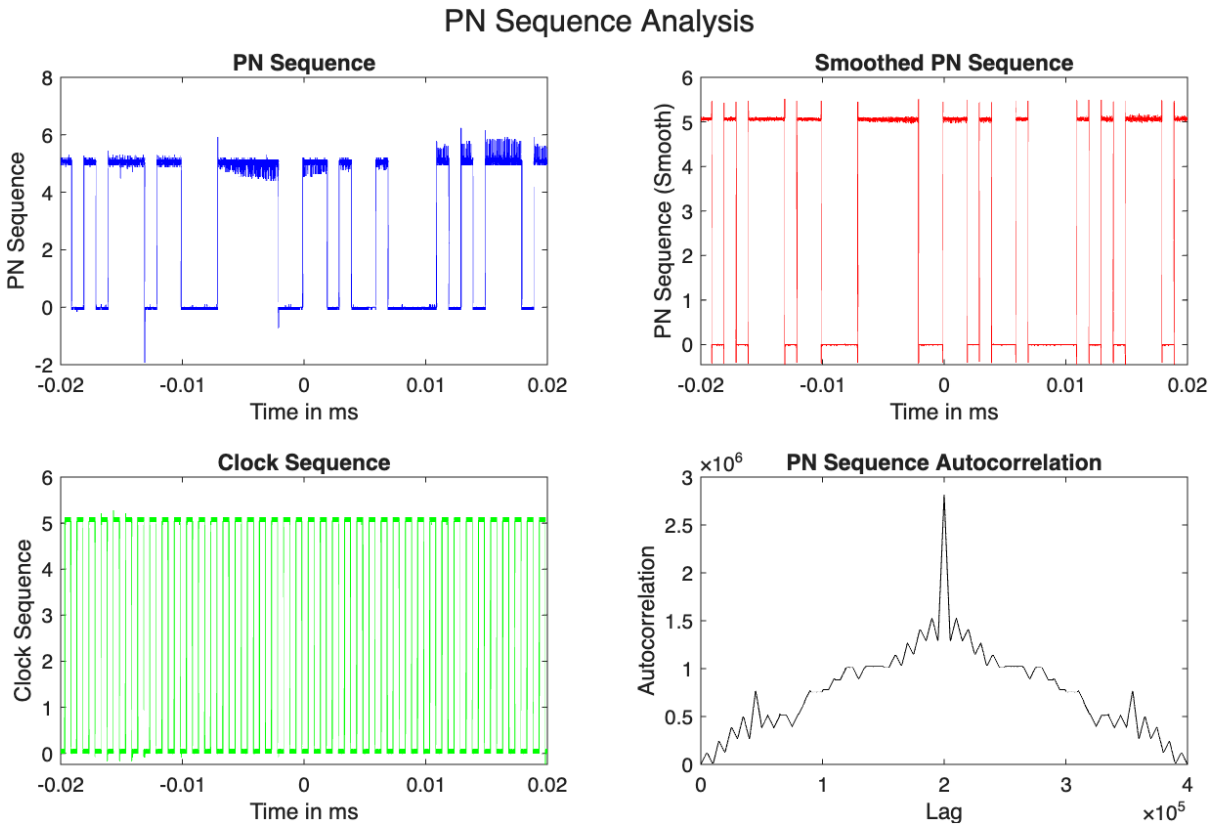


Figure 8: Plot of relevant PN Sequence data in MATLAB.

The following sequence plots the gold sequence, the clock pulse and its autocorrelation function using the data in .csv file.

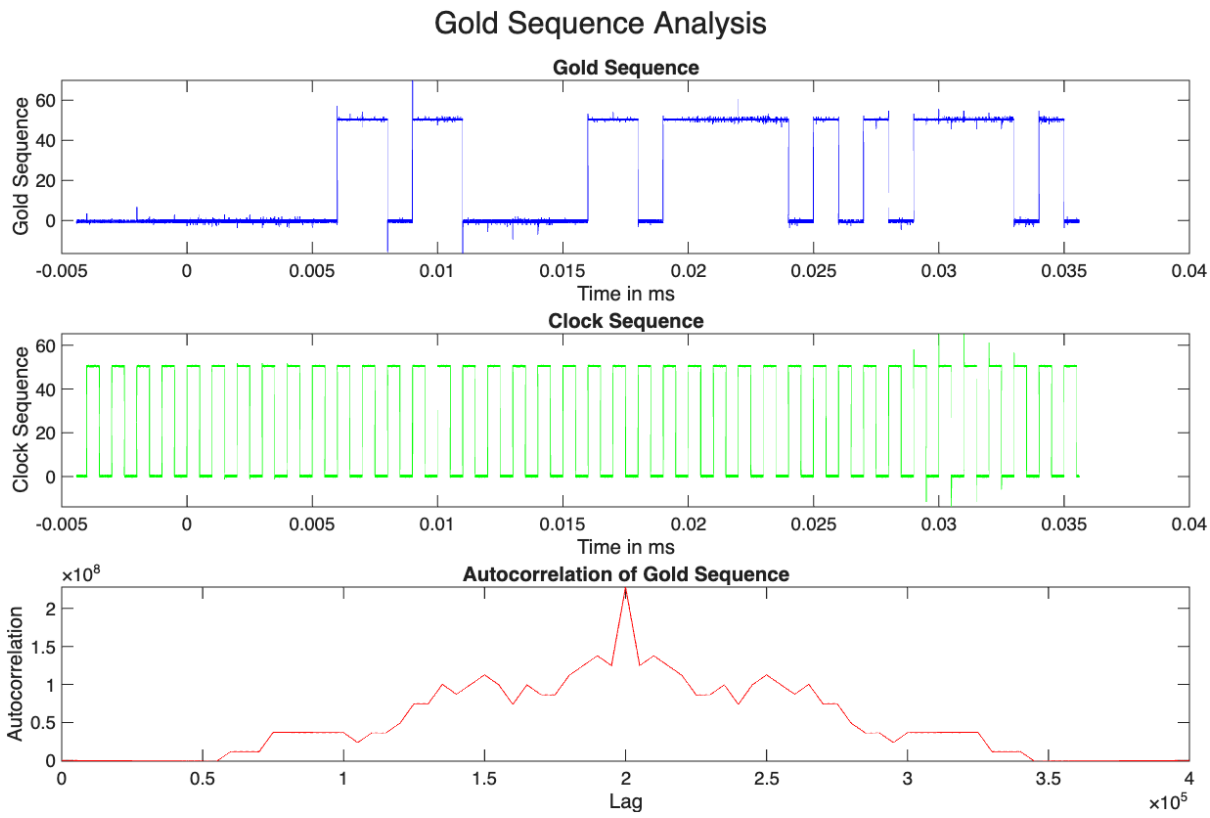


Figure 9: Plot of relevant Gold Sequence data in MATLAB.

2. Plot the autocorrelation function for the above data and comment on the generated function.

   The autocorrelation of the PN and Gold sequences are in Figure 8 and 9. Based on the plot, it can be inferred that the autocorrelation plot of the Gold sequence dies out early.

3. In addition, generate a PN sequence with 6-stage Linear Feedback Shift Register and plot its autocorrelation function using the following polynomial:

$$x^6 + x + 1$$

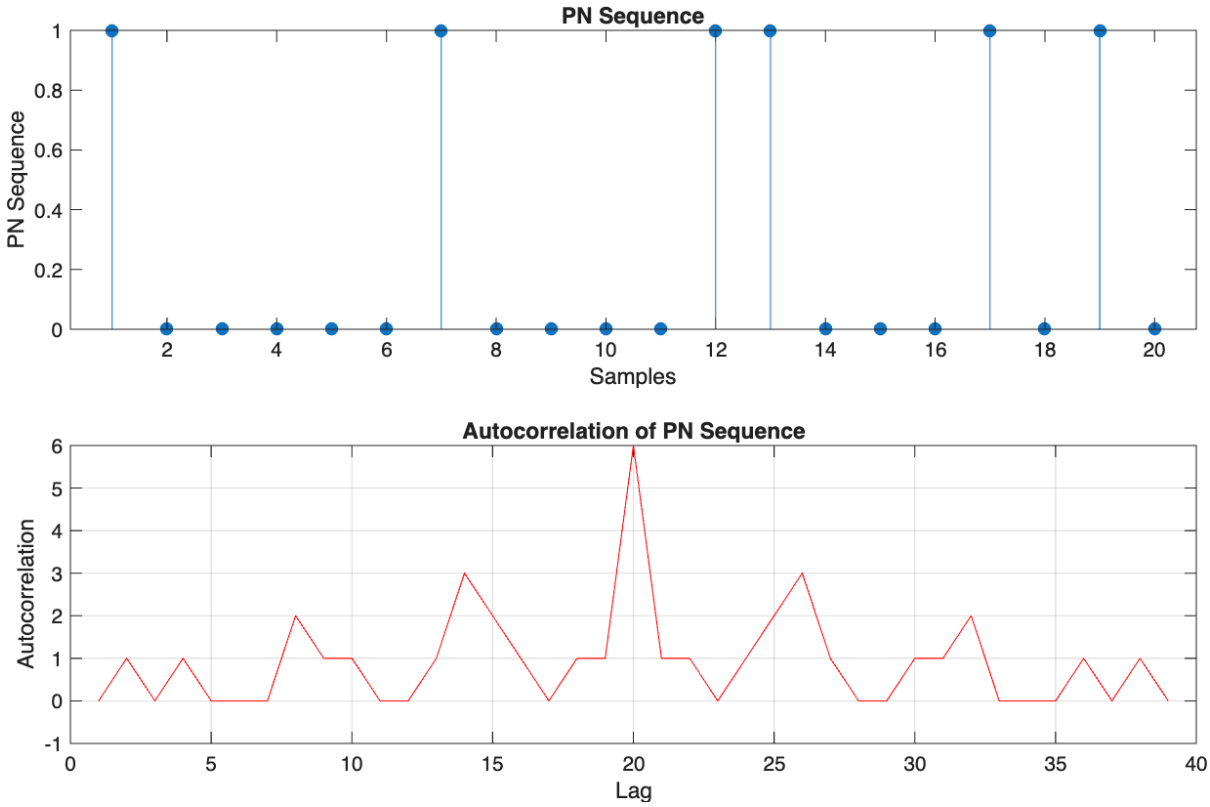


Figure 10: Plot of PN sequence and autocorrelation in MATLAB.

4. Repeat step 3 for the following polynomial:

$$x^6 + x^5 + x^4 + x^3 + x^2 + x + 1$$

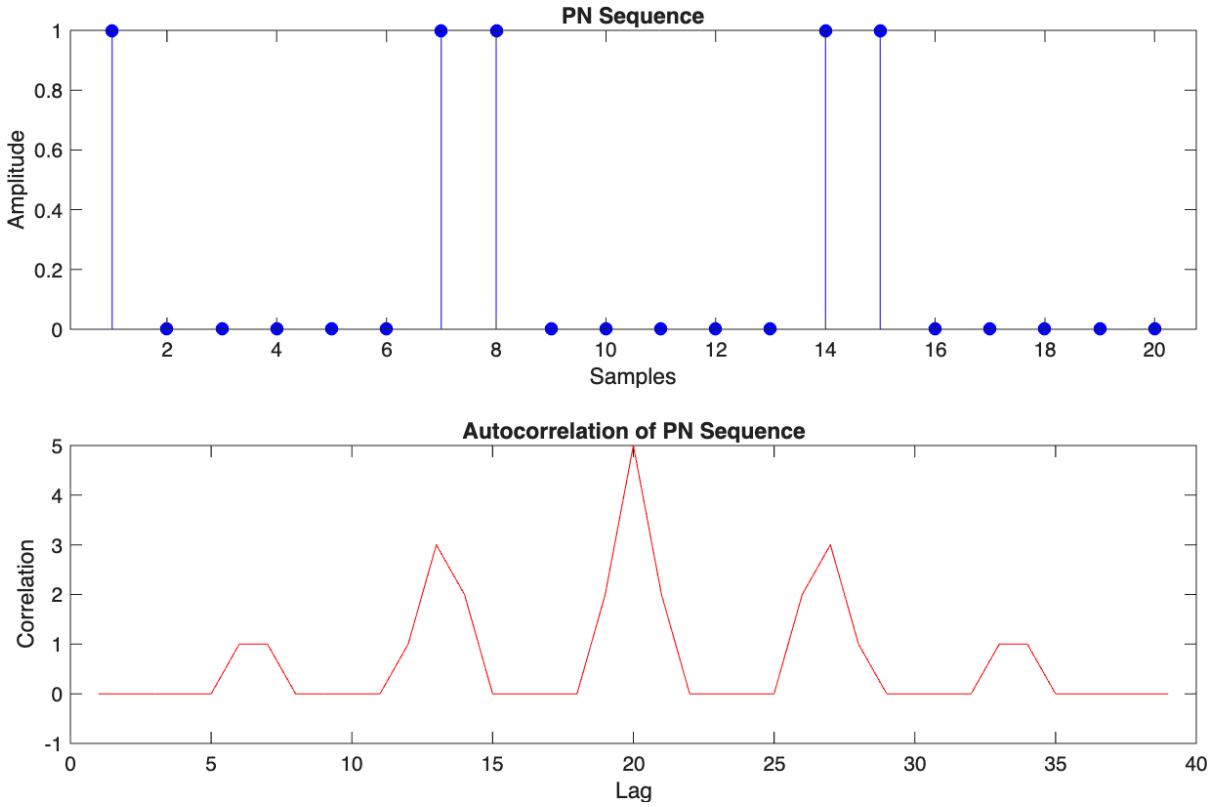


Figure 11: Plot of PN sequence and autocorrelation in MATLAB.

5. Comment on the difference in the plots, if any, emphasizing, in particular, on the nature of the polynomials (irreducible and primitive, if any).
   There are very few peaks when one uses multiple primitive polynomials.

APPENDIX:

Appendix: List of Primitive Polynomials

Degree	Primitive Polynomials
2	[2, 1, 0]
3	[3, 1, 0], [3, 2, 0]
4	[4, 1, 0], [4, 3, 0]
5	[5, 2, 0], [5, 3, 0], [5, 3, 2, 1, 0], [5, 4, 2, 1, 0], [5, 4, 3, 1, 0], [5, 4, 3, 2, 0]
6	[6, 1, 0], [6, 4, 3, 1, 0], [6, 5, 0], [6, 5, 2, 1, 0], [6, 5, 3, 2, 0], [6, 5, 4, 1, 0]
7	[7, 1, 0], [7, 3, 0], [7, 3, 2, 1, 0], [7, 4, 0], [7, 4, 3, 2, 0], [7, 5, 2, 1, 0]
7	[7, 5, 3, 1, 0], [7, 5, 4, 3, 0], [7, 5, 4, 3, 2, 1, 0], [7, 6, 0], [7, 6, 3, 1, 0]
7	[7, 6, 4, 1, 0], [7, 6, 4, 2, 0], [7, 6, 5, 2, 0], [7, 6, 5, 3, 2, 1, 0], [7, 6, 5, 4, 0]
7	[7, 6, 5, 4, 2, 1, 0], [7, 6, 5, 4, 3, 2, 0]
8	[8, 4, 3, 2, 0], [8, 5, 3, 1, 0], [8, 5, 3, 2, 0], [8, 6, 3, 2, 0], [8, 6, 4, 3, 2, 1, 0]
8	[8, 6, 5, 1, 0], [8, 6, 5, 2, 0], [8, 6, 5, 3, 0], [8, 6, 5, 4, 0], [8, 7, 2, 1, 0]
8	[8, 7, 3, 2, 0], [8, 7, 5, 3, 0], [8, 7, 6, 1, 0], [8, 7, 6, 3, 2, 1, 0]
8	[8, 7, 6, 5, 2, 1, 0], [8, 7, 6, 5, 4, 2, 0]

PIN DIAGRAMS:

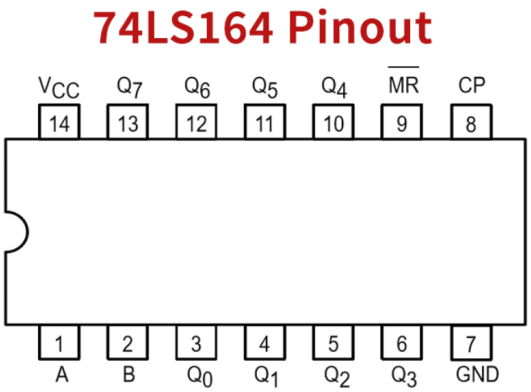


Figure 12: IC74164 [9]

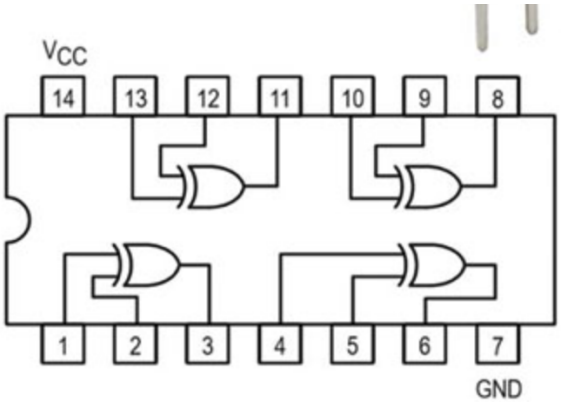


Figure 13: IC7486 [10]

DISCUSSION:
This was the first proper electronics experiment that I had. I was able to familiarize myself with the Digital Scope Oscillators, the electronic components, especially the ICs, the overall connections and the working of the electric components. My discussion with my lab partner was very fruitful as he helped me in understanding the overall working of the LFSRs and the PN sequence generators. I also relied on LLM models for better understanding of MATLAB codes that I referred in offical documentation. However, from my side there is still a need to understand the working of the GOLD sequence.

References

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(2) Bajaj, N. Nikeshbajaj/Linear_Feedback_Shift_Register: 1.0.6, version 1.0.6, 2021.

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(7) Jagganatham, A. K. CDMA Codes- Properties of PN Sequences, NPTEL Video Lecture, Available at https://nptel.ac.in/courses/117104115, 2025.

(8) Evans, B. L. Lab 4: Pseudo-Random Binary Sequences and Data Scramblers, Real-Time Digital Signal Processing Laboratory Course Material, ECE 445S, The University of Texas at Austin, Includes applications of PN sequences such as test signal generation, channel training, scramblers, and dither, 2025.

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(10) UTMEL 74LS86 Quad 2-Input XOR Gate IC: Pin, Datasheet PDF and Circuit Using 7486 Gate [FAQ], Accessed: 2025-08-21, 2021.